

SPT Preprocessing Follow-Up

*Validating Dynamic YAML Preprocessing Pipeline and
Investigating Timestep Threshold Cuts on SPT Data*

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- **Goal**

- Validate the yaml-configured BaseDataset path (PR #197) against the existing SPT baseline
- Study how timestep trimming changes the train/val phase space

- **Main question**

- What cuts isolate a *well-behaved* operating phase space without over-sculpting the training distribution?

- **PR #197 summary**

- Moves preprocessing choices into yaml: loader, transforms, split, scaling, NaN policy
- Existing subclass pipelines still work; the configured path is opt-in
- Makes it easy to audit which cuts define the *well-behaved* SPT phase space

- **Scan setup**

- Compare timestep-drop thresholds of 0.01, 0.02, 0.03, 0.04, 0.05, 0.10, 0.50, and 1.00
- Keep test minimally altered while cleaning only train/val; 1.00 is the no-cut baseline

Baseline YAML Layout

```
1  # SPT calibrator HDF5 data through the generic BaseDataset pipeline.
2
3  family: spt
4
5  loader:
6    type: spt_calibrator_hdf5
7    params:
8      root: /lcrp/project/SPT3G/users/ac.weiqian/ml_dqm/calibrator_responses
9      years: [2019, 2020, 2021, 2022, 2023]
10     data_variant: snr
11     response_template: calibrator_responses_095ghz_{year}.hdf5
12     snr_template: calibrator_response_snr_095ghz_{year}.hdf5
13     observation_id_key: Observation ID
14     require_monotonic_timestamps: true
15     load_response_reference: true
16     quality_reference_years: [2019, 2020, 2021, 2022, 2023]
17     wafer_id: w206
18     boloproperties_path: /lcrp/project/SPT3G/analysis/calarchive/v3/boloproperties/60000000.g3
19     feature_names:
20       response: response
21       snr: SNR
```

loader

- Dynamic YAML layout is easy to review at a glance.
- Validated against existing pipeline.

```
23 transforms:
24   after_load:
25     - name: select_stable_channels
26       fit: train
27       apply: [train, val, test]
28     params:
29       reference: response
30       quantiles: [10.0, 90.0]
31       tolerance: 0.10
32     - name: interpolate_nan_per_channel
33   post_split:
34     - name: filter_quality_timesteps
35       apply: [train, val]
36     params:
37       require_positive: true
38       quantiles: [1.0, 100.0]
39       min_valid_fraction: 1.0
```

transforms (after_load, post_split)

```
41 split:
42   mode: auto
43   strategy: by_year
44   ratio: [0.6, 0.2, 0.2]
45
46 scaling:
47   type: robust
48   mode: per_channel
49   source: train_set_fit
```

split & scaling

spt_configured.yml

Configured SPT Baseline Audit

split	type	shape	n_trainsteps	n_channels	n_features	n_stable	n_run	max_freq	min	max	rmax	std	rmedian	q51	q55	q55	q99
train	load	(12125, 303, 1)	12125	303	1	872935	4510	0.00040333	0	971.589	98.7940	38.1654	88.0887	0	31.2343	155.315	306.527
train	after_load_select_stable_channels	(12125, 113, 1)	12125	113	1	3620377	1348	0.808401526	0	971.589	97.8061	38.7474	94.2639	7.3352	47.1614	363.772	307.258
train	after_load_interpolate_run_per_channel	(12125, 113, 1)	12125	113	1	3630329	0	0	0	971.589	97.0885	38.7446	94.263	7.7364	47.1632	363.766	307.246
train	split	(19274, 113, 1)	18276	113	1	2177862	0	0	0	971.589	97.6886	38.9332	96.2635	6.87426	48.1396	363.805	308.894
train	post_split_filter_quality_timesteps	(12268, 113, 1)	12268	113	1	1186284	0	0	4.78185	971.589	122.831	38.8758	97.6338	35.9739	57.5532	365.977	311.425
train	scale	(12268, 113, 1)	12268	113	1	1186284	0	0	-2.71272	28.2845	0.19625	8.983683	-1.0338e-07	-5.20811	-0.923711	3.70807	8.85493
val	load	(12125, 303, 1)	12125	303	1	872935	4510	0.00040333	0	971.589	98.7940	38.1654	88.0887	0	31.2343	155.315	306.527
val	after_load_select_stable_channels	(12125, 113, 1)	12125	113	1	3620377	1348	0.808401526	0	971.589	97.8061	38.7474	94.2639	7.3352	47.1614	363.772	307.258
val	after_load_interpolate_run_per_channel	(12125, 113, 1)	12125	113	1	3630329	0	0	0	971.589	97.0885	38.7446	94.263	7.7364	47.1632	363.766	307.246
val	split	(6423, 113, 1)	6423	113	1	729786	0	0	0	760.77	97.8286	38.6687	96.2235	7.87145	47.1387	363.422	307.898
val	post_split_filter_quality_timesteps	(3813, 113, 1)	3813	113	1	430808	0	0	5.18954	760.77	123.435	34.6886	97.3832	36.8545	57.6463	364.894	311.446
val	scale	(3813, 113, 1)	3813	113	1	430808	0	0	-3.89999	14.1914	0.140588	8.989417	-0.89544368	-5.22285	-0.925889	3.71694	8.83532
test	load	(12125, 303, 1)	12125	303	1	872935	4510	0.00040333	0	971.589	98.7940	38.1654	88.0887	0	31.2343	155.315	306.527
test	after_load_select_stable_channels	(12125, 113, 1)	12125	113	1	3620377	1348	0.808401526	0	971.589	97.8061	38.7474	94.2639	7.3352	47.1614	363.772	307.258
test	after_load_interpolate_run_per_channel	(12125, 113, 1)	12125	113	1	3630329	0	0	0	971.589	97.0885	38.7446	94.263	7.7364	47.1632	363.766	307.246
test	split	(6428, 113, 1)	6428	113	1	729804	0	0	0	576.171	98.8986	38.1872	96.9297	11.4893	58.9832	162.35	288.11
test	scale	(6428, 113, 1)	6428	113	1	729804	0	0	-3.85225	15.6575	0.0348853	8.914089	-0.07748314	-0.38887	-0.98905	1.6349	2.83894

- This YAML configuration extracts the “Golden Dataset” with 113 stable channels (reduced from 303).
- Train/val timesteps after quality filtering: 19274/6423 → 12268/3813.
- Validated against the existing subclass method; confirmed 1:1 compatibility.

Timestep Scan: Config & Policy

```
19 transforms:
20   post_split:
21     - name: snr_threshold_nan
22       apply: [train, val]
23       params:
24         snr_min: 20.0
25     - name: drop_nan_timesteps
26       apply: [train, val]
27       params:
28         threshold: 0.2
29     - name: interpolate_nan_per_channel
30       apply: [train, val]
31       params:
32         require_all_filled: true
33     - name: fill_nan
34       apply: [test]
35       params:
36         value: 0.0
```

post_split timestep transforms

- Train/val logic

- $\text{SNR} < 20 \rightarrow \text{NaN}$
- Drop timesteps above the NaN-fraction threshold
- Interpolate remaining NaNs per channel
- Robust-scale with the train-fitted scaler

- Test logic

- No thresholding, dropping, or interpolation
- Fill NaN with 0, then apply the same train-fitted scaler

Only the train/val timestep-drop threshold changes across the scan; the test policy is held fixed.

[spt_timestep_trim.yml](#)

Timestep Trim Audit: 0.01 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_minib	n_max	nan_freq	min	max	mean	std	median	q01	q05	q95	q99
train	load	(32125, 2700, 1)	32125	2700	1	68713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.514	7.29552	45.4804	195.651	258.905
train	split	(16074, 2700, 1)	16074	2700	1	50911738	8962	0.00115492	0	1361.16	308.954	46.1422	302.356	5.74686	45.0626	194.806	258.822
train	post_split_err_threshold_nan	(16074, 2700, 1)	16074	2700	1	50905764	954036	0.0183338	20	1361.16	110.652	46.4804	302.276	32.391	55.7075	195.683	259.506
train	post_split_drop_nan_timesteps	(10740, 2700, 1)	10740	2700	1	28903976	94022	0.00324236	20.0003	1361.16	114.166	45.9384	306.233	39.5937	56.9918	198.385	262.255
train	post_split_drop_nan_per_channel	(10740, 2700, 1)	10740	2700	1	28998800	0	0	20.0003	1361.16	113.954	45.9807	306.088	39.0711	55.8028	198.045	262.119
train	scale	(10740, 2700, 1)	10740	2700	1	28998800	0	0	-3.75522	24.5125	0.149368	0.05777	0	-1.2237	-0.918881	1.72564	2.84641
val	load	(32125, 2700, 1)	32125	2700	1	68713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.514	7.29552	45.4804	195.651	258.905
val	split	(8425, 2700, 1)	8425	2700	1	17302910	1390	0.168406-05	0	1337.61	309.182	47.899	302.512	8.45186	45.4808	194.939	258.831
val	post_split_err_threshold_nan	(8425, 2700, 1)	8425	2700	1	17348501	291598	0.0168298	20	1337.61	110.912	46.6851	303.365	32.1631	54.7668	195.582	259.527
val	post_split_drop_nan_timesteps	(3430, 2700, 1)	3430	2700	1	9231036	29964	0.00323766	20.0011	1347.41	114.299	46.0825	306.384	39.4083	56.8002	198.551	262.903
val	post_split_drop_nan_per_channel	(3430, 2700, 1)	3430	2700	1	9241008	0	0	20.0011	1347.41	114.154	46.1227	306.269	38.9799	55.7057	198.424	262.809
val	scale	(3430, 2700, 1)	3430	2700	1	9241008	0	0	-3.47889	18.7266	0.153778	0.039384	0.0018757	-1.21381	-0.917865	1.73525	2.85377
test	load	(32125, 2700, 1)	32125	2700	1	68713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.514	7.29552	45.4804	195.651	258.905
test	split	(8426, 2700, 1)	8426	2700	1	17343812	34088	0.000840536	0	2139.13	309.779	47.9369	302.99	11.0486	48.8307	195.685	259.521
test	post_split_err_nan	(8426, 2700, 1)	8426	2700	1	17359400	0	0	0	2139.13	309.688	48.0222	302.948	9.65275	48.3095	195.574	259.487
test	scale	(8426, 2700, 1)	8426	2700	1	17359400	0	0	-5.59948	23.2178	0.0107699	0.037679	-0.074886	-0.86186	-1.54599	1.65436	2.88458

- Train/val kept: 10740 / 3430 (trimmed 44.3% / 46.6%).
- Aggressive cut: nearly half the train set removed; distribution shifts noticeably toward higher values.

Timestep Trim Audit: 0.02 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_finite	n_nan	nan_frac	min	max	mean	std	median	q01	q05	q95	q99
train	load	(32125, 2706, 1)	32125	2706	1	66713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
train	split	(15074, 2706, 1)	15074	2706	1	52931738	8862	0.000154902	0	1361.16	308.954	66.1422	302.356	5.76684	45.0426	194.866	258.822
train	post_split_trim_threshold_nan	(15074, 2706, 1)	15074	2706	1	52995764	954036	0.0183328	29	1361.16	110.052	46.4914	302.276	32.391	50.7075	195.683	259.506
train	post_split_drop_nan_timesteps	(14665, 2706, 1)	14665	2706	1	38356737	235793	0.00959305	29	1361.16	114.668	46.3818	306.653	39.4051	56.8365	199.713	264.451
train	post_split_interpolate_nan_per_channel	(14665, 2706, 1)	14665	2706	1	38396300	0	0	29	1361.16	114.424	46.4461	306.457	38.7044	55.9363	198.493	264.187
train	scale	(14665, 2706, 1)	14665	2706	1	38595530	0	0	-20.3623	72.5851	0.148871	0.889845	0	-1.23250	-0.918796	1.72883	2.85648
val	load	(32125, 2706, 1)	32125	2706	1	66713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
val	split	(6426, 2706, 1)	6426	2706	1	17609110	1980	0.00030405	0	1347.41	309.182	67.999	302.512	8.45386	45.4808	194.899	258.831
val	post_split_trim_threshold_nan	(6426, 2706, 1)	6426	2706	1	17648591	260598	0.0188298	29	1347.41	110.912	46.4461	303.365	32.1631	50.7668	195.582	259.527
val	post_split_drop_nan_timesteps	(4913, 2706, 1)	4913	2706	1	13161519	83561	0.0963685	20.0061	1347.41	114.524	46.4458	306.526	39.1312	55.8059	209.51	264.277
val	post_split_interpolate_nan_per_channel	(4913, 2706, 1)	4913	2706	1	13205300	0	0	20.0061	1347.41	114.284	46.4347	306.331	38.6439	55.3812	209.34	263.89
val	scale	(4913, 2706, 1)	4913	2706	1	13260300	0	0	-15.8788	37.0784	0.144463	0.876247	-0.8054824	-1.23347	-0.92253	1.72845	2.85267
test	load	(32125, 2706, 1)	32125	2706	1	66713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
test	split	(6426, 2706, 1)	6426	2706	1	17343812	14588	0.000840536	0	2139.13	306.779	47.9389	102.89	11.0496	46.8387	195.685	259.521
test	post_split_fit_mean	(6426, 2706, 1)	6426	2706	1	17039400	0	0	0	2139.13	309.688	48.0232	302.948	9.45273	48.3995	199.574	259.887
test	scale	(6426, 2706, 1)	6426	2706	1	17059400	0	0	-18.822	115.353	0.0508026	1.02934	-0.0802948	-2.46075	-1.15223	1.61682	2.84052

- Train/val kept: 14665 / 4913 (trimmed 23.9% / 23.5%).
- Heavy trim that recovers most of the lower-tail cleaning achieved at 0.01, with far more data retained.

Timestep Trim Audit: 0.03 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_finite	n_nan	nan_frac	min	max	mean	std	median	q01	q05	q95	q99
train	load	(32125, 2706, 1)	32125	2706	1	68713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
train	split	(14974, 2706, 1)	14974	2706	1	52931738	8862	0.000154902	0	1361.16	308.954	66.1422	302.356	5.74684	45.0426	194.866	258.822
train	post_split_trim_threshold_nan	(14974, 2706, 1)	14974	2706	1	52995764	954036	0.0183328	29	1361.16	110.052	46.4914	302.276	32.391	50.7075	195.683	259.596
train	post_split_trim_nan_timesteps	(14988, 2706, 1)	14988	2706	1	48211392	256208	0.00633119	29	1361.16	114.682	46.4584	306.679	39.3657	56.8366	199.786	264.523
train	post_split_interpolate_nan_per_channel	(14988, 2706, 1)	14988	2706	1	48867600	0	0	29	1361.16	114.448	46.4809	306.48	38.6812	55.9313	199.536	264.257
train	scale	(14988, 2706, 1)	14988	2706	1	48467600	0	0	-21.4124	75.1355	0.149723	0.87039	0	-1.23466	-0.82022	1.72834	2.85606
val	load	(32125, 2706, 1)	32125	2706	1	68713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
val	split	(6423, 2706, 1)	6423	2706	1	17460910	1980	0.00030605	0	1347.41	309.182	67.999	302.512	8.45386	45.4806	194.899	258.811
val	post_split_trim_threshold_nan	(6423, 2706, 1)	6423	2706	1	17468591	260598	0.0188298	29	1347.41	110.912	46.4851	303.365	32.1631	50.7868	195.582	259.527
val	post_split_trim_nan_timesteps	(5000, 2706, 1)	5000	2706	1	13418595	89465	0.00662259	20.0001	1347.41	134.57	46.4329	306.576	39.1317	55.8319	199.559	264.352
val	post_split_interpolate_nan_per_channel	(5000, 2706, 1)	5000	2706	1	13500800	0	0	20.0001	1347.41	114.341	46.4488	306.361	39.6415	55.41	199.392	264.88
val	scale	(5000, 2706, 1)	5000	2706	1	13500800	0	0	-16.2185	96.3852	0.143225	0.877967	-0.804756	-1.23829	-0.922857	1.72748	2.85387
test	load	(32125, 2706, 1)	32125	2706	1	68713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
test	split	(6426, 2706, 1)	6426	2706	1	17341812	14588	0.000840536	0	2139.13	309.779	47.9389	102.89	11.0496	46.8307	195.685	259.521
test	post_split_trim	(6426, 2706, 1)	6426	2706	1	17039400	0	0	0	2139.13	309.688	48.0232	302.948	9.45275	48.3995	199.574	259.887
test	scale	(6426, 2706, 1)	6426	2706	1	17059400	0	0	-15.586	117.324	0.0500289	1.02662	-0.0870080	-0.90266	-1.15268	1.61668	2.84308

- Train/val kept: 14988 / 5000 (trimmed 22.2% / 22.2%).
- Marginal relaxation from 0.02: similar data quality with a small number of extra timesteps admitted.

Timestep Trim Audit: 0.04 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_finfo	n_nan	nan_frac	min	max	mean	std	median	q01	q05	q95	q99
train	load	(32125, 2706, 1)	32125	2706	1	0.6713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.534	7.29552	45.4804	195.651	258.905
train	split	(15074, 2706, 1)	15074	2706	1	5.0911738	8962	0.00154902	0	1361.16	308.954	66.1422	302.366	5.74686	45.0626	194.806	258.822
train	post_split_err_threshold_nan	(15074, 2706, 1)	15074	2706	1	5.0905764	954036	0.0183338	20	1361.16	110.652	46.4804	302.276	32.391	55.7075	195.683	259.506
train	post_split_drop_nan_timesteps	(15550, 2706, 1)	15550	2706	1	4.9773114	313606	0.00747136	20	1361.16	114.85	46.434	306.117	36.0734	55.147	199.689	263.712
train	post_split_drop_nan_per_channel	(15550, 2706, 1)	15550	2706	1	4.9695800	0	0	20	1361.16	113.738	46.5282	305.871	36.824	54.9918	198.826	263.603
train	scale	(15550, 2706, 1)	15550	2706	1	4.9695800	0	0	-18.7965	65.932	0.144932	0.871247	0	-1.30487	-0.932737	1.72289	2.84663
val	load	(32125, 2706, 1)	32125	2706	1	0.6713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.534	7.29552	45.4804	195.651	258.905
val	split	(8425, 2706, 1)	8425	2706	1	1.7309110	1390	9.16836e-05	0	1347.41	309.182	47.999	302.532	8.45186	45.4808	194.939	258.833
val	post_split_err_threshold_nan	(8425, 2706, 1)	8425	2706	1	1.7348501	291598	0.0149298	20	1347.41	110.912	46.6851	303.365	32.1631	54.7968	195.582	259.527
val	post_split_drop_nan_timesteps	(5186, 2706, 1)	5186	2706	1	1.3994812	108108	0.0077265	20.9001	1347.41	114.027	46.4246	306.185	37.9638	55.9025	198.857	263.611
val	post_split_drop_nan_per_channel	(5186, 2706, 1)	5186	2706	1	1.4002200	0	0	20.9001	1347.41	113.731	46.409	305.067	37.0431	54.471	198.752	263.28
val	scale	(5186, 2706, 1)	5186	2706	1	1.4002200	0	0	-14.2384	51.3836	0.143816	0.870589	-0.8024363	-1.26169	-0.933223	1.72489	2.84604
test	load	(32125, 2706, 1)	32125	2706	1	0.6713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.534	7.29552	45.4804	195.651	258.905
test	split	(8426, 2706, 1)	8426	2706	1	1.7343812	34588	0.000840536	0	2139.13	309.779	47.9589	302.99	11.0486	48.8307	195.685	259.521
test	post_split_err_nan	(8426, 2706, 1)	8426	2706	1	1.7359400	0	0	0	2139.13	309.688	48.0222	302.948	9.49275	48.3995	195.574	259.407
test	scale	(8426, 2706, 1)	8426	2706	1	1.7359400	0	0	-27.7183	103.333	0.002876	1.00304	-0.6725458	-2.09191	-1.33318	1.94581	2.8582

- Train/val kept: 15550 / 5186 (trimmed 19.3% / 19.3%).
- Last threshold before a step-change in data quality; the lower tail is still well-behaved at this setting.

Timestep Trim Audit: 0.05 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_finfo	n_nan	nan_frac	min	max	mean	std	median	q01	q05	q95	q99
train	load	(32125, 2700, 1)	32125	2700	1	0.6713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.514	7.29552	45.4804	195.651	258.905
train	split	(16074, 2700, 1)	16074	2700	1	5.0911738	8962	0.00154902	0	1361.16	308.954	60.1422	302.356	5.74686	45.0626	194.806	258.822
train	post_split_err_threshold_nan	(16074, 2700, 1)	16074	2700	1	50305764	954036	0.0183338	20	1361.16	110.652	46.4804	302.276	32.391	50.7075	195.683	259.506
train	post_split_drop_nan_timesteps	(18115, 2700, 1)	18115	2700	1	48200860	626632	0.0128118	20	1361.16	111.573	46.4756	303.919	33.6086	51.7906	206.37	280.401
train	post_split_drop_nan_per_channel	(18115, 2700, 1)	18115	2700	1	68910900	0	0	20	1361.16	110.678	46.7732	303.636	29.5809	49.8713	195.922	258.87
train	scale	(18115, 2700, 1)	18115	2700	1	48910500	0	0	-6.76162	24.3821	0.118758	0.867681	0	-1.56480	-0.963911	1.40685	2.88313
val	load	(32125, 2700, 1)	32125	2700	1	0.6713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.514	7.29552	45.4804	195.651	258.905
val	split	(8423, 2700, 1)	8423	2700	1	17300910	1980	9.16860e-05	0	1347.61	309.182	47.899	302.512	8.65186	45.4808	194.899	258.833
val	post_split_err_threshold_nan	(8423, 2700, 1)	8423	2700	1	17048901	291598	0.0168298	20	1347.61	110.912	46.6851	303.365	32.1631	50.7668	195.582	259.527
val	post_split_drop_nan_timesteps	(6095, 2700, 1)	6095	2700	1	36237945	218455	0.0133355	20	1347.61	111.591	46.4487	303.962	33.4372	51.774	196.274	260.377
val	post_split_drop_nan_per_channel	(6095, 2700, 1)	6095	2700	1	38458300	0	0	20	1347.61	110.960	46.7337	303.488	29.5746	49.8915	195.848	259.795
val	scale	(6095, 2700, 1)	6095	2700	1	38458300	0	0	-5.43048	20.4886	0.128771	0.866938	-0.80133268	-1.54890	-0.961364	1.40959	2.88295
test	load	(32125, 2700, 1)	32125	2700	1	0.6713260	24240	0.000279464	0	2139.13	189.16	40.0736	302.514	7.29552	45.4804	195.651	258.905
test	split	(8426, 2700, 1)	8426	2700	1	17343812	34588	0.000840536	0	2139.13	309.779	47.9369	162.99	11.0486	48.8307	195.685	259.521
test	post_split_err_nan	(8426, 2700, 1)	8426	2700	1	17059400	0	0	0	2139.13	309.688	48.0222	302.948	9.69275	48.3995	195.574	259.407
test	scale	(8426, 2700, 1)	8426	2700	1	17059400	0	0	-9.87338	37.1885	0.0850248	0.917044	-0.0229113	-0.08960	-1.85906	1.40862	2.87627

- Train/val kept: 18115 / 6095 (trimmed 6.0% / 5.1%).
- Crossing this boundary admits a visible drop in lower-tail quality, with around 2500 extra train timesteps appearing.

Timestep Trim Audit: 0.10 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_finite	n_nan	nan_frac	min	max	mean	std	median	q01	q05	q95	q99
train	load	(10125, 2706, 1)	32125	2706	1	66713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
train	split	(10174, 2706, 1)	10174	2706	1	50931738	8862	0.000154902	0	1361.16	308.954	66.1422	302.356	5.76684	45.0426	194.806	258.822
train	post_split_trim_threshold_nan	(10174, 2706, 1)	10174	2706	1	52095764	954036	0.0183328	29	1361.16	110.052	46.4914	303.276	32.391	50.7075	195.683	259.506
train	post_split_drop_nan_timesteps	(10881, 2706, 1)	10881	2706	1	56405569	763131	0.0148907	29	1361.16	110.064	46.4722	303.389	32.7091	51.8136	195.723	259.734
train	post_split_interpolate_nan_per_channel	(10881, 2706, 1)	10881	2706	1	512468700	0	0	29	1361.16	110.18	46.8234	302.894	26.0478	48.8311	195.187	259.114
train	scale	(10881, 2706, 1)	10881	2706	1	512468700	0	0	-8.82218	24.4241	0.127059	0.885085	0	-1.54740	-0.969599	1.68862	2.88003
val	load	(10125, 2706, 1)	32125	2706	1	66713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
val	split	(6423, 2706, 1)	6423	2706	1	17360910	1980	0.000305	0	1147.41	309.182	67.999	302.512	8.45386	45.4806	194.899	258.811
val	post_split_trim_threshold_nan	(6423, 2706, 1)	6423	2706	1	17368591	260598	0.0168298	29	1147.41	110.912	66.4851	303.365	32.1631	50.7668	195.582	259.527
val	post_split_drop_nan_timesteps	(6386, 2706, 1)	6386	2706	1	16977700	264402	0.0133396	29	1147.41	111.048	46.4589	303.478	32.4963	51.0532	195.689	259.646
val	post_split_interpolate_nan_per_channel	(6386, 2706, 1)	6386	2706	1	17242200	0	0	29	1147.41	110.247	46.793	302.901	28.853	48.8744	195.186	259.027
val	scale	(6386, 2706, 1)	6386	2706	1	17242200	0	0	-8.89705	20.4363	0.128063	0.869901	-0.00379627	-1.54686	-0.969766	1.69029	2.88128
test	load	(10125, 2706, 1)	32125	2706	1	66713260	24240	0.000279464	0	2139.13	109.16	46.0736	302.514	7.29952	45.4804	195.651	258.905
test	split	(6426, 2706, 1)	6426	2706	1	17343812	14588	0.000840536	0	2139.13	309.779	47.9369	102.89	11.0496	46.8307	195.685	259.521
test	post_split_fit_norm	(6426, 2706, 1)	6426	2706	1	17039600	0	0	0	2139.13	309.688	48.0232	302.948	9.45275	48.3995	196.574	259.887
test	scale	(6426, 2706, 1)	6426	2706	1	17059600	0	0	-8.89029	23.108	0.112268	0.910188	0.00194103	-1.96077	-1.0418	1.69086	2.88198

- Train/val kept: 18981 / 6386 (trimmed 1.5% / 0.6%).
- Very light trim: only about 1.5% of train timesteps removed; distribution is nearly identical to the no-cut baseline.

Timestep Trim Audit: 0.50 Threshold

split	step	shape	n_timesteps	n_channels	n_features	n_freq	n_nsn	nns_bac	min	max	mean	std	median	q01	q05	q95	q99	
train	load	(132125, 2700, 1)	32125	2700	1	68713260	24240	0.000279464	0	2139.13	189.16	48.0736	302.514	7.29552	45.4804	195.651	258.905	
train	split	(16074, 2700, 1)	16074	2700	1	520321780	8862	0.000154902	0	1361.16	308.054	48.1422	302.354	5.76484	45.0426	194.804	258.822	
train	post_split_snz_threshold_nsn	(16074, 2700, 1)	16074	2700	1	50305764	954036	0.0183328	20	1361.16	110.052	46.4814	302.276	32.391	50.7015	195.683	259.596	
train	post_split_drop_nsn_timesteps	(15045, 2700, 1)	15045	2700	1	50773497	886033	0.0178902	20	1361.16	110.058	46.4873	302.28	32.4369	50.8096	195.694	259.596	
train	post_split_outspeculate_nsn_per_channel	(15045, 2700, 1)	15045	2700	1	50960300	0	0	20	1361.16	110.011	46.0388	302.681	27.1491	48.5523	195.629	258.925	
train	scale	(16074, 2700, 1)	16045	2700	1	50960300	0	0	-6.26633	24.5642	0.126031	0.865881	0	-1.55352	-1.80385	1.68719	2.87605	
val	load	(32125, 2700, 1)	32125	2700	1	68713260	24240	0.000279464	0	2139.13	189.16	48.0736	302.514	7.29552	45.4804	195.651	258.905	
val	split	(6425, 2700, 1)	6425	2700	1	17340910	1990	9.16836e-05	0	1347.41	309.142	47.999	302.512	8.45386	45.4808	194.999	258.831	
val	post_split_snz_threshold_nsn	(6425, 2700, 1)	6425	2700	1	17048501	280598	0.0148098	20	1347.41	110.912	46.4851	302.365	32.1431	50.7468	195.582	259.527	
val	post_split_drop_nsn_timesteps	(6418, 2700, 1)	6418	2700	1	17043093	285407	0.0164703	20	1347.41	110.927	46.4794	302.376	32.2271	50.7018	195.591	259.536	
val	post_split_outspeculate_nsn_per_channel	(6418, 2700, 1)	6418	2700	1	17328800	0	0	0	20	1347.41	110.071	46.0343	302.787	27.6735	48.4814	195.652	258.878
val	scale	(6418, 2700, 1)	6418	2700	1	17328800	0	0	-6.12085	20.4886	0.128097	0.866688	-0.00025916	-1.58274	-0.986888	1.68926	2.87405	
test	load	(132125, 2700, 1)	32125	2700	1	68713260	24240	0.000279464	0	2139.13	189.16	48.0736	302.514	7.29552	45.4804	195.651	258.905	
test	split	(6426, 2700, 1)	6426	2700	1	17343812	14588	0.00040536	0	2139.13	309.779	47.9389	302.93	11.0496	48.6007	195.685	259.521	
test	post_split_fit_nsn	(6426, 2700, 1)	6426	2700	1	17053800	0	0	0	2139.13	309.684	48.0222	302.948	9.45275	48.3995	195.374	259.487	
test	scale	(6426, 2700, 1)	6426	2700	1	17053800	0	0	-6.86075	30.2812	0.114897	0.900089	0.0052004	-1.58316	-1.20794	1.68973	2.88332	

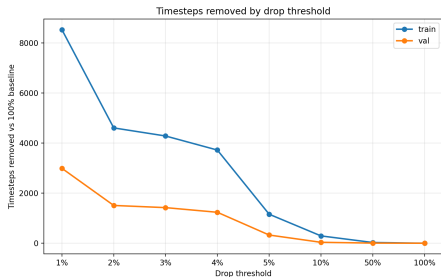
- Train/val kept: 19245 / 6418 (trimmed 0.2% / 0.1%).
- Effectively no timestep trimming: the distribution matches the SNR-only baseline.

Timestep Trim Audit: 1.00 Threshold

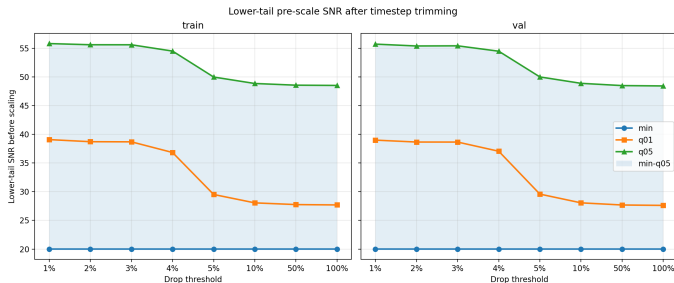
split	step	shape	n_timesteps	n_channels	n_features	n_finite	n_nan	nan_frac	min	max	mean	std	median	q01	q05	q95	q99
train	load	(19275, 2706, 1)	32125	2706	1	68713260	24240	0.000279464	0	2139.13	199.16	46.0736	302.514	7.29552	45.4804	195.651	258.905
train	split	(19274, 2706, 1)	19274	2706	1	52931738	8862	0.000154902	0	1361.16	308.954	46.1422	302.356	5.74684	45.0426	194.806	258.822
train	post_split_trim_threshold_nan	(19274, 2706, 1)	19274	2706	1	52995764	954036	0.0183328	29	1361.16	110.052	46.4814	303.278	32.391	50.7075	195.683	259.596
train	post_split_drop_nan_timesteps	(19274, 2706, 1)	19274	2706	1	52995764	954036	0.0183328	29	1361.16	110.052	46.4814	303.278	32.391	50.7075	195.683	259.596
train	post_split_interpolate_nan_per_channel	(19274, 2706, 1)	19274	2706	1	52939800	0	0	29	1361.16	309.089	46.0811	302.674	27.4885	48.5013	194.96	258.89
train	scale	(19274, 2706, 1)	19274	2706	1	52939800	0	0	-8.26014	24.5582	0.125809	0.866062	0	-1.55632	-1.09471	1.60896	2.87055
val	load	(19275, 2706, 1)	32125	2706	1	68713260	24240	0.000279464	0	2139.13	189.16	46.0736	302.514	7.29552	45.4804	195.651	258.905
val	split	(6425, 2706, 1)	6425	2706	1	17340910	1390	0.0000805	0	1147.41	309.182	47.999	302.512	8.05386	45.4806	194.899	258.811
val	post_split_trim_threshold_nan	(6424, 2706, 1)	6423	2706	1	17348591	260598	0.0189298	29	1147.41	110.912	46.4851	303.365	32.1631	50.7068	195.582	259.527
val	post_split_drop_nan_timesteps	(6425, 2706, 1)	6423	2706	1	17348591	260598	0.0189298	29	1147.41	110.912	46.4851	303.365	32.1631	50.7068	195.582	259.527
val	post_split_interpolate_nan_per_channel	(6425, 2706, 1)	6423	2706	1	17342100	0	0	29	1147.41	110.044	46.0411	302.768	27.6067	48.4182	195.03	258.852
val	scale	(6425, 2706, 1)	6423	2706	1	17342100	0	0	-8.13072	20.4897	0.128391	0.867071	-0.002802807	-1.55736	-0.968761	1.60816	2.87928
test	load	(19275, 2706, 1)	32125	2706	1	68713260	24240	0.000279464	0	2139.13	189.16	46.0736	302.514	7.29552	45.4804	195.651	258.905
test	split	(6426, 2706, 1)	6426	2706	1	17341812	14088	0.00040536	0	2139.13	309.779	47.9389	102.89	11.0496	46.8307	195.685	259.521
test	post_split_fit_norm	(6426, 2706, 1)	6426	2706	1	17339400	0	0	0	2139.13	309.688	48.0232	302.948	9.05275	48.3995	196.574	259.887
test	scale	(6426, 2706, 1)	6426	2706	1	17355600	0	0	-8.94723	32.2908	0.114865	0.890217	0.0055565	-1.98112	-1.03757	1.60925	2.88384

- Train/val kept: 19274 / 6423 (trimmed 0.0% / 0.0%).
- No timestep trimming applied; serves as the reference baseline for comparison across the scan.

Threshold Scan Summary



timesteps removed vs threshold



lower-tail quantile movement

- Timesteps removed drop sharply below a 5% threshold; anything looser keeps nearly all data.
- A clear quality break occurs between 4% and 5%: crossing that line admits around 2500 extra train timesteps with noticeably worse lower-tail behavior.
- 4% appears to be the cleanest practical cut: tight enough to remove poor-quality data without discarding too much of the training set.

- **Baseline pipeline behaves as expected.** Stable-channel filtering reduces 303 channels to 113, and quality filtering trims the training set substantially before any threshold scan.
- **The scan mainly changes the lower tail.** As the trim gets stricter, the lowest-signal timesteps are removed while the mean, median, and upper end of the distribution move much less.
- **There is a sharp quality break between 4% and 5%.** Crossing that line admits around 2500 more train timesteps that visibly worsen the low-signal tail of the distribution.
- **1%–4% are the tight-cleaning regime.** These thresholds remove 20–44% of train timesteps and keep the data distribution the most well-behaved.
- **5%–100% are comparatively mild.** These settings keep nearly all timesteps but admit the poorer-quality lower tail.
- **Recommendation.** 4% looks like the best candidate cut: it sits just before the quality drop and avoids over-sculpting the training set.